

Anaesthesia and Surgery

Malignant hyperthermia (RYR1, CACNA1S), butyrylcholinesterase (BCHE), MC1R, postoperative nausea (CHRM3), anesthetic and analgesic metabolism, perioperative thrombosis and healing

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Key variants by topic	6
5. Favourable	8
6. Unfavourable	9
7. Mechanisms of ageing	10
8. Practical suggestions	11
9. Limitations	12
10. Methods and sources	13
11. Annex: all variants analysed	14

1. Summary

Favourable. No variants of malignant hyperthermia were detected: RYR1 G2434R and CACNA1S R1086H normal here, and 34 + 2 SNPs without variants in the profile. The risk is reduced, but not absent. Butyrylcholinesterase (BCHE) is normal, without the atypical variant or K, so succinylcholine and mivacurium have normal duration. There are no **MC1R** variants of red hair, so the need for volatile anesthetics and the response to local anesthetics should be normal. The CYP2D6 is probable normal (*1/*2): ondansetron, codeine and tramadol work as expected. CYP2B6 does not have *6 (ketamine and propofol). Factor V Leiden and prothrombin G20210A are negative and G6PD is normal.

Unfavourable. The venous thrombosis score is at percentile 86.3 (the same sign of \$49), relevant for prophylaxis after surgery. CYP2C9 is *1/*3 (perioperative NSAIDs with more exposure). With **VKORC1 TT**, Sérgio F is very sensitive to warfarin. It has 1 allele of **CHRM3** associated with postoperative nausea and vomiting (small effect). **OPRM1 A/G** and **COMT Val/Val** request a little more opioid. **UGT1A1 *1/*28** (Gilbert) can increase bilirubin with fasting, which is benign.

What does it mean? Sérgio F genetics does not show anesthetic contraindications. What counts most in surgery are the drugs that take: LDN (blocks opioids), rapamycin (delays healing), dapagliflozin (ketoacidosis), retatrutide (slow gastric emptying, aspiration risk), isotretinoin (cicatrication), tadalafil and omega-3. These are associated with the prophylaxis of thrombosis by the high score. All this should be told to the anesthesiologist and the surgeon before any surgery.

Polygenic scores update (03-10-2026). The scores were recalculated with more published recent scores (the same method and the same reference population). This box prevails over the previous values cited in the text.

- Venous thrombosis: 3 new scores give p24 and p71 (genomic) and p80 (10 SNPs); with the previous ones (p86, p84 and p54), the consensus is **p76**. It goes from high to above average; the practical rules remain.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 13 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 11 of the chip, 1 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 0 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 1 PGS Catalog scores calculated locally with 265 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

A high percentile = more risk of venous thrombosis. There are no validated polygenic scores for the response to anesthesia; this score enters perioperative prophylaxis.

Perioperative thrombosis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Venous thrombosis	PGS000043 (Klarin 2019, 297 variants)	265 297 (weighted coverage 84%)	86.3	MORE LIKELY	unfavourable

- **Venous Thromboembolism:** Already seen in §26 and §49; repeated here by perioperative prophylaxis.

4. Key variants by topic

Variants with recommendations in anesthesia (malignant hyperthermia, butyrylcholinesterase) and variants that change the choice or dose of perioperative drugs.

Malignant hyperthermia

Variants of RYR1 and CACNA1S can cause malignant hyperthermia with volatile anesthetics and succinylcholine: a rare but severe crisis, with muscle stiffness, fever and acidosis.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
RYR1	rs118192172	R614C	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	T faux copy/s)	without data	R614C: pathogenic variant of malignant hyperthermia (triggers: volatile anesthetics and succinylcholine)	Strong
RYR1	rs121918592	G2434R	G/G	chip	A (0 copy/s)	favourable	G2434R: pathogenic variant of malignant hyperthermia	Strong
CACNA1S	rs1800559	R1086H	C/C	chip	T (0 copy/s)	favourable	R1086H: pathogenic variant of malignant hyperthermia	Strong

Muscle relaxants

Butyrylcholinesterase (BCHE) degrades succinylcholine and mivacurium. With low activity variants, paralysis and apnea can last for hours.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
BCHE	rs1799807	D70G (atypical)	T/T	chip	C (0 copy/s)	favourable	Asp70Gly (atypical variant, C in strand +): slow butyrylcholinesterase; prolonged apnea with succinylcholine and mivacurium in homozygous	Strong
BCHE	rs1803274	A539T (K)	C/C	chip	T (0 copy/s)	favourable	Ala539Thr (variant K, T in strand +): ~30% less activity; only relevant in homozygous or combined with atypical	Moderate

Anaesthetic sensitivity and nausea

MC1R alters the need for anesthetics and pain; CHRM3 is associated with postoperative nausea and vomiting.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
MC1R	rs1805007	R151C	C/C	chip	T (0 copy/s)	favourable	R151C (red hair): more need for volatile anesthetics (~20%), more sensitivity to thermal pain and less effect of local anesthetics	Moderate
MC1R	rs1805008	R160W	C/C	chip	T (0 copy/s)	favourable	R160W (red hair): same effect of R151C, especially in carriers of 2 alleles	Moderate
CHRM3	rs2165870		A/G	imputation DR2=0.98	A (1 copy/s)	unfavourable (light)	Allele A is associated with postoperative nausea and vomiting (GWAS; OR ~1.2-1.4	Weak

Perioperative drug metabolism

Enzymes that metabolize analgesics, antiemetics and anticoagulants used around the surgery, and factor V (thrombosis).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CYP2D6	rs3892097	*4	C/C	chip	T (0 copy/s)	favourable	*4 (null): ondansetron, codeine, tramadol, metoprolol	Strong
CYP2C9	rs1057910	*3	A/C	chip	C (1 copy/s)	unfavourable	*3: Perioperative NSAIDs (ibuprofen, parecoxib/celecoxib, ketorolac) with more exposure	Strong
CYP2B6	rs3745274	*6	G/G	chip	T (0 copy/s)	favourable	*6: more exposed ketamine, methadone and propofol	Strong
UGT1A1	rs887829	*28 (proxy)	C/T	chip	T (1 copy/s)	mixed	Marks *28 (Gilbert): bilirubin rises with fasting and surgical stress (benign)	Strong
F5	rs6025	Leiden	C/C	chip	T (0 copy/s)	favourable	Factor V Leiden: perioperative venous thrombosis ↑ (detail in §49)	Strong

5. Favourable

- **Malignant hyperthermia:** without the variants RYR1 G2434R and CACNA1S R1086H; in the profile, 34 SNPs of RYR1 and 2 of CACNA1S without variants. Reduced risk, but not absent: the chip does not read all variants of RYR1 (R614C was not read). Strong for the tested variants.
- **Normal BCHE** (without the atypical variant D70G or K A539T): succinylcholine and mivacurium with normal duration. **Strong.**
- **MC1R without R151C or R160W:** need for volatile anesthetics and response to local anesthetics expected as normal. **Moderate.**
- **Probable normal CYP2D6** (*1/*2; no copy number): effective ondansetron; codeine and tramadol normally activated. Strong for alleles tested.
- **CYP2B6 without *6:** ketamine without increased exposure. **Moderate.**
- **No hereditary thrombophilia detected:** factor V Leiden C/C, prothrombin G20210A negative (§49). Normal G6PD. **Strong.**

6. Unfavourable

- **Venous thrombosis:** score at percentile 86.3 (297 variants; §26 and §49). It justifies mechanical and/or pharmacological prophylaxis according to the type of surgery, and early mobilization. **Moderate.**
- **CYP2C9 *1/*3:** ibuprofen, celecoxib/parecoxib, meloxicam and piroxicam with more exposure. Use the minimum dose and avoid long half-life NSAIDs. **Strong.**
- **CYP2C9 *1/*3 + VKORC1 TT:** very sensitive to warfarin (if it is necessary to anticoagulate after surgery, heparin with low molecular coefficients or a direct anticoagulant avoid the problem; if it is warfarin, dose guided by the genotype). **Strong.**
- **CHRM3 rs2165870 A/G:** 1 allele of nausea and postoperative vomiting (OR ~1.2-1.4 by allele). **Weak.**
- **OPRM1 A/G and COMT Val/Val:** need for a slightly higher opioid. **Weak moderate.**
- **UGT1A1 *1/*28 (Gilbert):** bilirubin can rise with fasting and surgical stress (mild and benign jaundice is not liver injury). Strong for the genotype, without gravity.

7. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Vascular Aging	Venous thrombosis at percentile 86.3; surgery and immobilization add risk	Unfavourable (conditional)
Chronic Inflammation	Surgical trauma generates an acute inflammatory response; omega-3 and glucose control attenuate it	Conditional
Remodelling of the extracellular matrix	Wound healing: rapamycin and isotretinoin delay collagen formation and wound closure	Unfavourable (by drugs)
Nutrient Sensing Deregulation	The inhibition of mTOR by rapamycin reduces the proliferation of fibroblasts and healing	Conditional
Fragility	Postoperative recovery depends on previous strength and mobility; pre-habilitation (training before surgery) reduces complications	Neutral

8. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **To deliver to the anesthesiologist a leaf with pharmacogenetics:** CYP2C9 *1/*3, VKORC1 TT, CYP2D6 *1/*2 probable (without copy number), CYP2B6 *2/*4, UGT1A1 *1/*28, BCHE normal, RYR1/CACNA1S without detected variants (not excluding malignant hyperthermia), OPRM1 A/G and high thrombosis score. *Suggestion (not applied).*
2. **LDN 4.5 mg:** suspend 24–72 hours before surgery in which opioids are expected, and only resume when no longer necessary. *Suggestion (not applied).*
3. **Rapamycin 4 mg every 6 weeks:** planning surgery away from a dose. Do not take since ~1–2 weeks before the wound is healed (decide with the surgeon). *Suggestion (not applied).*
4. **Dapagliflozin 10 mg:** suspend 3 days before surgery (risk of euglycemic ketoacidosis) and resume when you return to eat and drink normally. *Suggestion (not applied).*
5. **Retatrutida (class GLP-1):** delays gastric emptying and increases the risk of aspiration in anesthesia. Inform the anesthesiologist and follow the guidance of suspension or liquid diet before surgery. *Suggestion (not applied).*
6. **Acarbose and imeglimin:** not to take on the day of fasting. Tadalafil: no nitrates in the 48 hours after the last dose (inform the team). *Suggestion (not applied).*
7. **Isotretinoin:** in elective surgeries and aesthetic procedures (laser, dermoabrasion, plastic surgery), discuss with the dermatologist and the surgeon pause and waiting time, because of healing. *Suggestion (not applied).*
8. **Omega-3 3.2–4 g/day:** inform the surgeon; in high risk surgeries hemorrhagic can request a short break. *Suggestion (not applied).*
9. **Prophylaxis of thrombosis (high percentile score):** compression or pneumatic compression, heparin with low molecular coefficients if the type of surgery justifies it, and rise and walk on the day itself or the following day. Long flights after surgery, hydration and socks. *Suggestion (not applied).*
10. **Postoperative analgesia:** paracetamol (maximum 2–3 g/day while the ALT is high) and NSAID of short duration in the minimum dose (CYP2C9 *1/*3). If necessary opioid: dose adjusted to response (OPRM1 A/G) and LDN suspended. *Suggestion (not applied).*
11. **Postoperative nausea:** ask for prophylaxis with ondansetron (effective with normal CYP2D6) and dexamethasone. *Suggestion (not applied).*
12. **Pre-habilitation:** maintain strength and aerobic training until surgery, adequate protein and controlled glucose. It reduces complications and accelerates recovery. *Suggestion (not applied).*

9. Limitations

- The chip reads only some variants of RYR1 and CACNA1S (there are more than 50 pathogenics). The absence of variants does not exclude malignant hyperthermia; the family history of anesthetic complications remains essential.
- CYP2D6 was read by SNPs; deletions and duplications of the gene were not evaluated.
- The venous thrombosis score has 297 variants and 84% of coverage of the coefficients; it is a relative risk sign, not absolute.
- There are no validated polygenic scores for healing or postoperative nausea; CHRM3 has a small effect.
- The recommendations on drugs come from anesthesia guidelines, not genetics; the final decision is from the anesthesiologist and the surgeon.
- Conclusions only from genes; it is not a diagnosis.

10. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the imputed VCF; in the palindromic SNPs the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic score:** Harmonized GRCh38 file from PGS Catalog (vein thrombosis, Klarin 2019, 297 variants). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Interpretation:** International pharmacogenetic guidelines for RYR1/CACNA1S (Gonsalves 2019), CYP2C9 and NSAIDs (Theken 2020), CYP2C9/VKORC1 and warfarin (Johnson 2017) and CYP2D6 with ondansetron and opioids (Bell 2017; 2021 crews); Liu 2005 for MC1R; Klenke 2018 for CHRM3; anesthesia guidelines for SGLT2, GLP-1 and mTOR. GWAS Catalog and ClinVar. The complete references are available on request.

11. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Malignant hyperthermia	RYR1	rs118192172	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	T			SNP not found in the data
Malignant hyperthermia	RYR1	rs121918592	G/G	chip	A	0	homozygous for the reference allele (monomorphic site on chip)	without the risk allele
Malignant hyperthermia	CACNA1S	rs1800559	C/C	chip	T	0	homozygous for the reference allele (monomorphic site on chip)	without the risk allele
Muscle relaxants	BCHE	rs1799807	T/T	chip	C	0	strand +	without the risk allele
Muscle relaxants	BCHE	rs1803274	C/C	chip	T	0	strand +	without the risk allele
Anaesthetic sensitivity and nausea	MC1R	rs1805007	C/C	chip	T	0	strand +	without the risk allele
Anaesthetic sensitivity and nausea	MC1R	rs1805008	C/C	chip	T	0	strand +	without the risk allele
Anaesthetic sensitivity and nausea	CHRM3	rs2165870	A/G	imputation DR2=0.98	A	1	strand +	1 copy of risk allele
Perioperative drug metabolism	CYP2D6	rs3892097	C/C	chip	T	0	strand +	without 4: Probable normal CYP2D6 (1/*2; no copy number)
Perioperative drug metabolism	CYP2C9	rs1057910	A/C	chip	C	1	strand +	1/3: intermediate metabolizer
Perioperative drug metabolism	CYP2B6	rs3745274	G/G	chip	T	0	strand +	without *6
Perioperative drug metabolism	UGT1A1	rs887829	C/T	chip	T	1	strand +	1/28: bilirubin can rise with perioperative fasting (benign)
Perioperative drug metabolism	F5	rs6025	C/C	chip	T	0	strand +	without the risk allele